

# BIOLOGY EXAM-ORIENTED STUDY FORMAT

## Strategic High-Weightage Analysis Checklist

Created for Thunderstudy By Pragya

### The "Golden List": High-Probability Topics by Unit

Use this list as your primary revision checklist. These topics cover approximately 70–80% of the paper's weightage.

#### Unit VI: Reproduction (High Weightage for Diagrams & Reasoning)

- **Sexual Reproduction in Flowering Plants:** focus on **Mutualism** (Fig Tree & Wasp), **Sexual Deceit** (Ophrys), and Embryo Sac structure.
- **Human Reproduction:** **Menstrual Cycle** (master the LH/FSH/Progesterone graph), Gametogenesis, and the neuroendocrine mechanism of Parturition.
- **Reproductive Health:** ART (IVF, ZIFT, GIFT, ICSI) and Contraception modes (IUDs, Pills).

#### Unit VII: Genetics & Evolution (High Weightage for Case Studies)

- **Genetics:** Disorders (Down's, Turner's, Klinefelter's), **Pedigree Analysis** (Haemophilia), and Molecular Basis (Nucleosome, Lac Operon, Replication Fork).
- **Evolution:** Natural Selection graphs (Stabilizing, Directional, Disruptive), Industrial Melanism, and Homologous vs Analogous organs.

#### Unit VIII: Biology in Human Welfare (Easy Scoring)

- **Human Health:** Malaria life cycle, Immunity (Antibody structure), and Drugs (Opioids, Cannabinoids).
- **Microbes:** **Sewage Treatment (STP)** – Primary/Secondary, Flocs, Activated Sludge, and BOD.

## Unit IX: Biotechnology (Technical & Process-Oriented)

- **Principles:** **EcoRI** naming and action, **pBR322 Vector** (memorize restriction sites), and PCR steps.
- **Applications:** Bt Cotton mechanism, RNA interference (RNAi), and Insulin maturation.

## Unit X: Ecology (Graph-based)

- **Organisms & Populations:** Logistic Growth Curve (Verhulst-Pearl) and Age Pyramids.
- **Biodiversity:** Species–Area Relationship (graph and equation) and the "**Evil Quartet**".

### The "Sure-Shot" Questions (Repeated Verbatim)

1. **Sewage Treatment:** Explain biological treatment, flocs, activated sludge, and BOD changes.
2. **Restriction Enzymes:** Explain naming of EcoRI and its action showing sticky ends.
3. **Cloning Vector:** Draw pBR322; explain selectable markers and Insertional Inactivation.
4. **Mutualism:** Fig tree vs Wasp OR Orchid vs Bee relationship.
5. **Bioreactors:** Simple vs Sparged Stirred-tank difference.

## Exam-Oriented Diagrams to Practice

- **pBR322 Vector** (Label AmpR, TetR, ori, rop)
- **Sewage Treatment Plant (STP)** Flow Chart
- **Human Sperm** (Acrosome, Mitochondria, Tail)
- **Nucleosome** (Histone octamer, DNA wrap)
- **Menstrual Cycle Graph** (Hormonal peaks)
- **Anatropous Ovule** & Mature Embryo Sac

## Revision Strategy

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### Day 1: Diagram Mastery

Draw the core diagrams listed above until perfect. A good diagram is 50% of the mark.

### Day 2: Flowcharts

Map out Malaria Life Cycle, Central Dogma, STP, MOET, and PCR.

### Day 3: High Weightage Theory

Focus solely on the "Golden List" topics. Skip low-weightage details.

### Day 4: Case Studies

Practice reading growth graphs and Pedigree charts.

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**Biology Exam Strategy By Pragya Thunderstudy**  
**Analyze. Revise. Dominate.**